

LAWRENCEVILLE-VINCENNES, IL, USA

WMO: 725342

Lat: 38.764N

Lon: 87.606W

Elev: 429

StdP: 14.47

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 13809

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.6	11.2	-3.2	4.7	8.5	1.2	5.9	12.2	28.3	35.2	26.0	40.7	7.2	320	0.493

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.1	93.7	75.6	91.4	75.1	89.7	74.4	79.5	89.5	78.1	87.7	76.8	85.6	9.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.7	141.4	85.1	75.2	134.2	83.5	73.8	128.0	81.5	43.2	89.5	41.7	87.6	40.3	85.4	86.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.7	21.0	18.8	DB	-2.1	97.3	5.1	3.5	-5.8	99.8	-8.8	101.9	-11.7	103.8	-15.4	106.4	
			WB	-2.8	82.1	4.9	1.8	-6.4	83.4	-9.3	84.4	-12.0	85.4	-15.6	86.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	56.1	31.2	35.3	45.0	56.6	66.7	74.8	77.3	76.1	70.0	57.8	45.5	35.2
	DBStd	18.60	11.44	10.89	10.65	8.99	8.11	5.85	5.08	4.91	6.92	9.11	9.36	10.64
	HDD50	1970	589	424	226	38	1	0	0	0	0	27	195	470
	HDD65	4671	1047	833	623	276	82	6	0	1	31	262	587	923
	CDD50	4181	7	12	71	236	518	743	846	808	600	269	59	12
	CDD65	1407	0	0	4	24	134	299	381	344	181	39	1	0
	CDH74	13407	0	1	31	233	1173	2837	3799	3192	1738	392	11	0
	CDH80	5337	0	0	3	38	389	1162	1632	1302	695	115	1	0
Wind	WSAvg	8.4	10.2	10.0	10.3	10.3	8.7	7.2	6.0	5.7	6.3	8.0	9.0	9.2
Precipitation	PrecAvg	43.5	2.7	2.4	3.9	4.1	5.2	4.0	4.3	3.8	3.0	3.1	4.0	3.3
	PrecMax	60.1	8.1	5.4	10.7	9.7	12.4	8.7	10.9	11.8	8.5	9.7	11.2	8.1
	PrecMin	31.7	0.5	0.5	0.7	1.0	1.3	0.4	0.2	0.6	0.0	0.2	0.1	0.6
	PrecStd	6.6	1.8	1.3	2.2	2.1	2.9	2.0	2.5	2.2	1.9	2.2	2.1	1.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.9	68.7	78.6	84.1	90.9	95.1	98.0	97.0	94.9	88.4	74.9	65.9
		MCWB	59.8	60.0	64.0	64.3	71.4	74.5	78.8	77.3	72.6	70.7	61.2	60.7
	2%	DB	58.9	62.4	72.4	80.5	87.9	92.0	93.5	92.9	91.1	82.8	70.0	60.4
		MCWB	54.5	55.4	60.4	65.0	70.7	74.6	77.3	75.8	72.7	67.2	60.1	56.9
	5%	DB	54.0	57.1	67.3	76.3	84.4	90.1	90.9	90.1	88.0	78.6	64.8	55.0
		MCWB	50.9	51.5	57.6	62.4	69.1	73.9	76.9	75.4	71.8	64.7	57.6	51.1
	10%	DB	48.0	52.1	62.8	72.3	81.2	87.3	88.4	87.4	83.8	73.2	61.1	50.7
		MCWB	44.1	46.9	54.9	60.7	68.1	72.7	75.4	74.2	69.2	62.6	54.0	46.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.5	61.9	65.4	70.3	76.1	79.3	81.8	81.0	77.1	73.2	66.2	62.0
		MCDB	63.2	67.5	76.4	79.4	85.1	89.9	91.5	90.4	88.4	84.1	72.3	65.2
	2%	WB	55.6	57.3	62.4	67.3	73.6	77.5	80.1	78.9	75.2	70.4	62.1	57.1
		MCDB	58.2	60.8	70.3	76.0	82.6	87.6	90.1	88.6	85.7	79.0	66.7	59.6
	5%	WB	51.2	52.4	59.4	64.9	71.8	76.0	78.6	77.4	73.6	67.4	58.9	51.8
		MCDB	53.5	56.6	65.9	72.2	80.7	85.7	88.2	86.6	83.7	74.8	64.6	54.4
	10%	WB	44.1	46.4	55.8	62.7	69.9	74.7	77.4	75.9	71.9	64.5	54.7	47.0
		MCDB	47.2	50.7	61.7	69.7	77.8	83.6	86.2	84.0	80.1	71.4	60.2	49.6
Mean Daily Temperature Range	5% DB	MDBR	15.7	16.7	19.0	20.6	19.4	19.6	19.1	20.2	23.2	21.9	18.7	14.8
		MCDBR	20.8	24.0	25.2	25.6	23.5	22.9	21.4	23.1	26.5	26.8	24.6	19.6
		MCWBR	17.9	18.6	15.6	13.5	9.9	8.9	8.7	9.8	10.3	13.0	16.3	16.4
	5% WB	MCDBR	19.8	21.8	22.3	20.6	19.5	19.2	19.1	19.5	22.0	22.8	21.3	17.5
MCWBR		18.5	18.7	15.1	13.4	9.7	8.7	9.3	9.8	10.0	12.1	16.6	16.1	
Clear-Sky Solar Irradiance	taub	0.312	0.324	0.360	0.409	0.445	0.449	0.468	0.475	0.423	0.362	0.323	0.308	
	taud	2.448	2.394	2.365	2.229	2.179	2.199	2.149	2.116	2.251	2.422	2.476	2.466	
	Ebn at Noon	274	285	285	275	266	265	258	253	260	268	267	267	
	Edh at Noon	26	31	36	43	47	46	48	48	40	30	25	24	
All-Sky Solar Radiation	RadAvg	600	846	1178	1547	1742	2009	1942	1776	1475	1061	716	514	
	RadStd	68	93	107	92	153	128	122	102	106	111	77	50	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page